

AGE 316: MICROCOMPUTER SYSTEMS

LECTURE 11: LIBRARIES & FUNCTIONS

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Recap: Advanced Interfacing

Displays: We connected LCDs and OLEDs.

I2C Protocol: We saved pins using SDA/SCL.

Code: We used `LiquidCrystal_I2C`... but what *is* that?

> **Today:** We learn how to write **Clean, Modular Code** using Functions and Libraries.

The "Spaghetti Code" Problem

Messy Code

As your projects grow, `void loop()` becomes huge.

- Hard to read.
- Hard to debug.
- Copy-pasting the same code 10 times? Bad idea!

```
void loop() {
  digitalWrite(2, HIGH);
  delay(100);
  digitalWrite(2, LOW);
  delay(100);
  // ... repeat 50 times ...
  sensorRead = analogRead(A0);
  // ... complicated math ...
}
```

Solution: Functions

A **Function** is a named block of code that performs a specific task.



Encapsulate

Wrap complex logic into a single named command.



Reuse

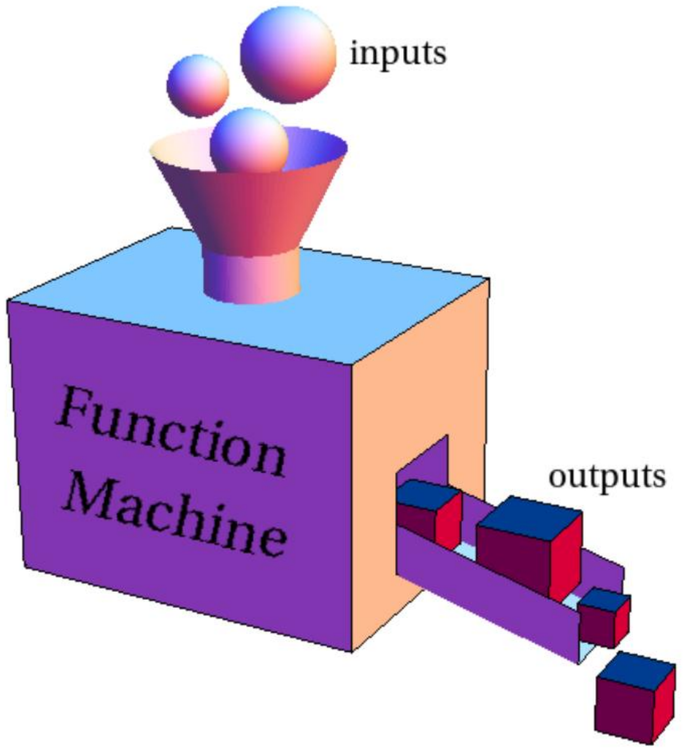
Write once, call many times.



Organize

Keep your main loop clean and readable.

The "Function Machine"



Inputs & Outputs

Think of a function as a machine/factory.

1. **Input (Parameters):** Raw materials (Numbers, Text).
2. **Processing (Body):** The work happens inside the box.
3. **Output (Return):** The finished product.

Anatomy of a Function

How to define a function in C++:



ReturnType: What comes out? (`int`, `float`, `void`).

Name: Descriptive action verb (e.g., `blinkLED`).

Parameters: Inputs needed (e.g., `int pin`).

Type 1: Void Functions

"Doers"

These functions do an action but give nothing back.

Keyword: `void` (Empty/Nothing).

Example: Turning on an LED, printing to Serial.

```
// Definition
void blinkRed() {
  digitalWrite(9, HIGH);
  delay(500);
  digitalWrite(9, LOW);
}

// Usage in loop
blinkRed();
```

Type 2: Return Functions

"Calculators"

These functions perform a calculation and send the result back.

Keyword: `return`.

Example: Converting Celsius to Fahrenheit.

```
// Definition
float getVoltage(int pin) {
  int val = analogRead(pin);
  float v = val * (5.0 / 1023.0);
  return v;
}

// Usage
float myVolts = getVoltage(A0);
```

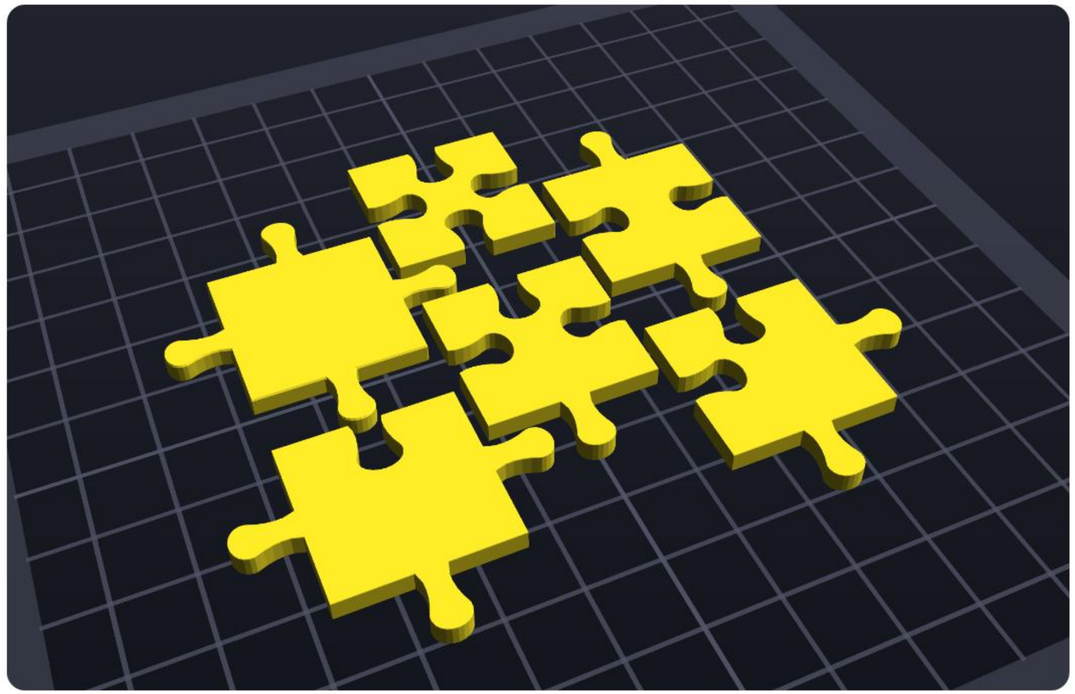
Parameters: Passing Data

Make your functions flexible by passing arguments.

```
// Flexible Blinker
void blink(int pin, int time) {
  digitalWrite(pin, HIGH);
  delay(time);
  digitalWrite(pin, LOW);
  delay(time);
}

// Call it with different data
blink(13, 100); // Fast blink on 13
blink(9, 1000); // Slow blink on 9
```

Moving Up: Libraries



"Don't Reinvent the Wheel"

A **Library** is a collection of functions written by someone else to solve common problems.

- Controlling Servos? `Servo.h`
- Driving LCDs? `LiquidCrystal.h`
- Math? `Math.h`

Anatomy of a Library

A library usually consists of two files:



.h (Header)

The "Menu". Lists all the functions available for you to use.



.cpp (Source)

The "Kitchen". Contains the actual code and logic (hidden from you).

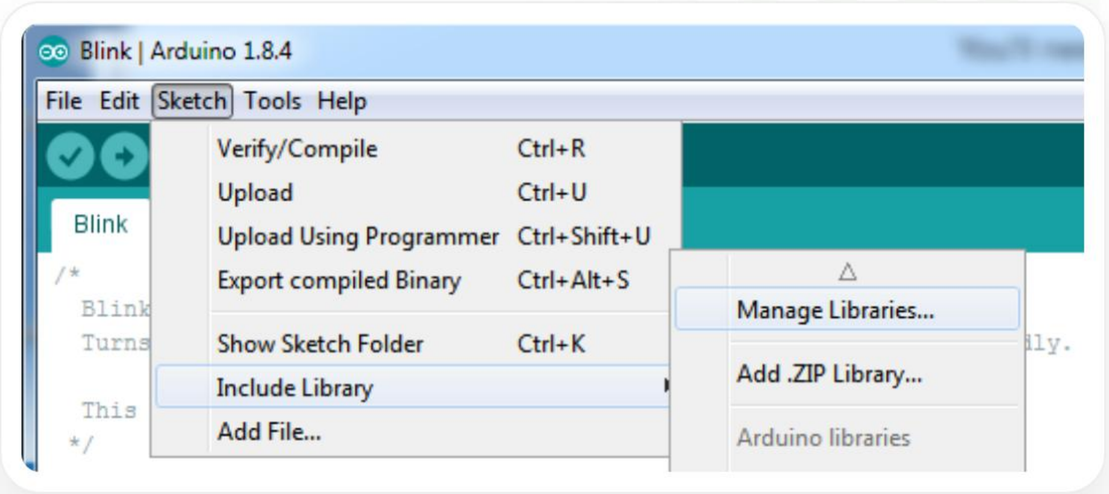
You include them using: `#include`

Installing Libraries

Library Manager

The easiest way. It's like an App Store for code.

1. Sketch → Include Library → Manage Libraries.
2. Search for the name.
3. Click Install.



Using Libraries (Objects)

Most libraries require you to create an **Object** (Instance).

```
#include // 1. Include

Servo myMotor; // 2. Create Object

void setup() {
  myMotor.attach(9); // 3. Use Object methods
}
```

Think of `Servo` as the blueprint, and `myMotor` as the actual robot you built.

Engineering Ethics: Clean Code

Code is read by humans more often than by machines.

A

Naming

`int x` is bad.

`int motorSpeed` is good.

Small Functions

A function should do **one thing** only.

Comments

Explain *Why*, not just *What*.

Practical: Modular Farm Monitor

```
void loop() {
  float t = readTemp();
  float h = readHumidity();

  if (t > 30) {
    activateCooling();
  }

  updateDisplay(t, h);
  delay(1000);
}
```

Look how clean that loop is! All the complexity is hidden in functions.

Summary



Functions

Modular blocks of code. Use `void` or return types.



Libraries

Pre-written code packages. Install via Library Manager.



Clean Code

Write code that is easy to read and maintain.

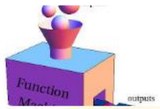
Questions?

"The only stupid question is the one you don't ask."

Dr. Kadeghe Fue

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Image Sources



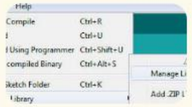
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